



Department of Electrical and Electronic Engineering

Jashore University of Science and Technology

PROJECT REPORT

Course Code: EEE 3200

Name of the Project: *CardioPredictor: A Reliable and Non-Invasive Health Monitoring with Cardiovascular Disease Prediction System using Random Forest Classifier*

Remarks:

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Submitted by:

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Year: 3rd
Semester: 2nd

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Objectives:

- ☐ To measure non-invasive blood glucose, cholesterol, cuffless systolic and diastolic blood pressure.
- ☐ To monitor the real-time data on webpage.
- ☐ To predict the existence of cardiovascular disease based on data.

Abstract:

CardioPredictor is a bloodless and cuffless health monitoring system designed to measure blood glucose, cholesterol, systolic, and diastolic blood pressure. By placing fingers on the sensors, the real-time health data can be displayed on an OLED screen as well as on Thingspeak server through Wi-Fi within 30 seconds. A web-based application then retrieves the latest data and combines it with user inputs (age, height, weight, lifestyle factors) to predict cardiovascular disease risk using a machine learning based model (Random Forest) with Approximately 72% accuracy. The entire process is completed in under one minute, providing users with timely and actionable health insights.

Introduction:

Health monitoring plays a vital role in maintaining a healthy lifestyle and preventing the onset of non-communicable diseases (NCDs) such as diabetes, hypertension, and cardiovascular diseases. These conditions contribute significantly to global mortality rates, necessitating regular health checks to detect early signs of declining health. The rapid development of technology has enabled the creation of innovative electronic devices that monitor key health parameters in a non-invasive and cuffless systems to maintain their well-being.

Blood glucose is a critical parameter for diagnosing and managing diabetes. Normal fasting blood glucose levels range between 70–100 mg/dL, while levels above 126 mg/dL indicate diabetes. Low blood glucose (hypoglycemia) can cause symptoms like dizziness, confusion, and seizures, while high levels (hyperglycemia) can lead to severe complications such as organ damage and diabetic ketoacidosis. Traditional glucose measurement methods are painful and inconvenient, involving finger pricks or blood samples. Non-invasive glucose monitoring provides a painless and user-friendly alternative for effective diabetes management. (al, 2022)

Cholesterol is another essential health indicator, with normal total cholesterol levels being below 200 mg/dL. Levels between 200–239 mg/dL are considered borderline high, while levels above 240 mg/dL are categorized as high cholesterol. High cholesterol levels can result in plaque buildup in arteries, increasing the risk of heart attacks and strokes. Conversely, excessively low cholesterol levels may lead to hormonal imbalances and other health issues. Regular monitoring of cholesterol levels, especially through non-invasive methods, helps maintain a healthy lipid profile and prevents cardiovascular risks. (al, 2022)

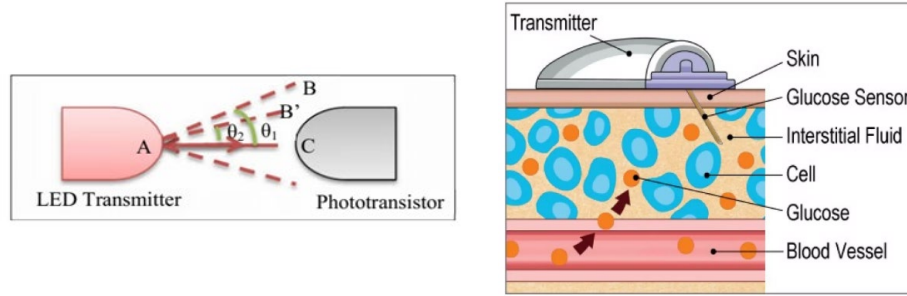


Fig 1: Non-Invasive method of measuring glucose

To measure non-invasive blood glucose and cholesterol levels, an empirical equation is used. The input, **X**, represents the average IR value recorded over 30 seconds by the MAX30101 sensor's **940 nm photodetector**. Infrared radiation penetrates the skin and interacts with glucose molecules in the blood. The remaining IR light is reflected and detected by the photodetector. This detected IR value is then used to estimate glucose and cholesterol levels. (Usman Umar, 2020)

$$\text{Glucose (mg/dL)} = -0.0011X + 194.87 \dots \dots \dots (\text{eq 1})$$

$$\text{Cholesterol (mg/dL)} = -0.0033X + 531.49 \dots \dots \dots (\text{eq 2})$$

Blood pressure, measured in terms of systolic (SBP) and diastolic (DBP) pressures, is crucial for identifying hypertension. Normal blood pressure is below 120/80 mmHg, while elevated levels above 140/90 mmHg indicate hypertension. High blood pressure can lead to complications like heart disease, kidney damage, and stroke, while extremely low blood pressure (hypotension) can cause fainting and shock due to inadequate blood flow. Traditional methods like sphygmomanometers are accurate but often bulky or require professional assistance. Non-invasive and cuffless devices using Photoplethysmogram (PPG) signals provide a practical and convenient alternative for long-term monitoring. (M. Bunkum, 2021)

In a cuffless blood pressure monitoring system, two sensors are placed at different locations on the body to measure ECG and PPG signals. The time difference between the ECG R-peak and the corresponding peak in the PPG waveform, known as **Pulse Transit Time (PTT)**, is used to estimate **systolic (SBP) and diastolic (DBP) blood pressure** using an empirical equation below, where Td is the Heart Rate (HR) in ms – Time delay; and Time delay is HR(bpm)+ PTT.

$$\text{SBP (mmHg)} = 184.3 - 1.329 \times \text{HR(bpm)} + 0.00848 \times \text{Td} \dots \dots \dots (\text{eq 3})$$

$$\text{DBP (mmHg)} = 55.96 - 0.02912 \times \text{HR(bpm)} + 0.02302 \times \text{Td} \dots \dots \dots (\text{eq 4})$$

(M. Bunkum, 2021)

In this research, a machine learning-based **Random Forest model** with an accuracy of approximately 72% is employed to predict cardiovascular disease risk. By combining real-time sensor data for glucose, cholesterol, SBP, and DBP with user-provided details such as age, weight, height, smoking, alcohol and daily activity level, the system delivers timely and actionable health

insights. Regular monitoring combined with predictive analytics ensures early detection of potential health risks and empowers users to take preventive measures effectively.

Components Required:

Table 1: Required components for the project

SL No.	Component Name	Rating/Specifications	Quantity
01	SparkFun Particle Sensor Breakout – MAX30101	Red(660nm), IR(940nm) and Green(537nm) LEDs. Operating Voltage:5V	1
02	SKU SEN0203 Heart Pulse Sensor	Operating Voltage: 3.3V – 5V Current Consumption: <4mA Output: Analog signal Detection Range: 30–240 BPM LED: Green (525nm)	1
03	ESP8266 NodeMCU board	Operating Voltage: 3.3V Input Voltage: 4.5V – 9V (via Vin pin) or 5V via USB Flash Memory: 4MB (ESP-12E module) WiFi Frequency: 2.4 GHz	1
04	OLED Display Blue I2C 128x32 0.91 inch	Display Type: OLED (Organic Light Emitting Diode) Resolution: 128 × 32 pixels Screen Size: 0.91 inches Color: Blue	
05	Veroboard	Dot type	1
06	USB type B cable	5V- 3A	1
07	Connecting wires	-	As required

Table 2: Required data for cardiovascular prediction system

SL No.	Data	Specification
01	Glucose Level	mg/dl
02	Cholesterol Level	mg/dl
03	Systolic Blood Prssure (SBP)	mmHg
04	Diastolic Blood Prssure (DBP)	mmHg
05	Age	days
06	Height	cm
07	Weight	Kg
08	Gender	0 = male; 1 = female
09	Smoking habit	0 = No; 1= Yes
10	Alcohol intake	0 = No; 1= Yes
11	Physical Activity	0 = inactive; 1= Active

Circuit Diagram:

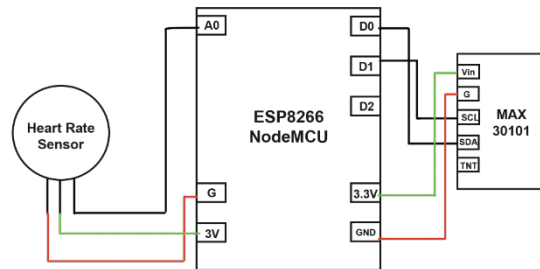


Fig 2: Connection diagram of CardioPredictor device

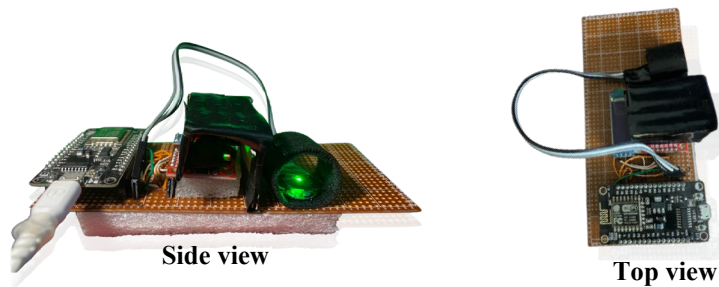


Fig 3: Physical view of the project

Methodology:

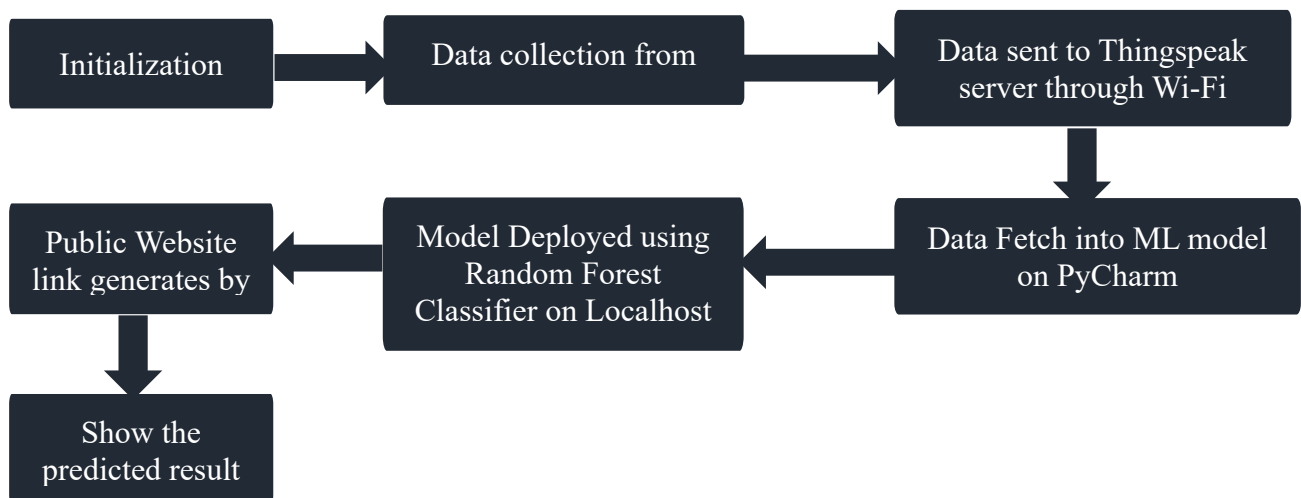


Fig 4: Methodology from developer end

Flow Chart:

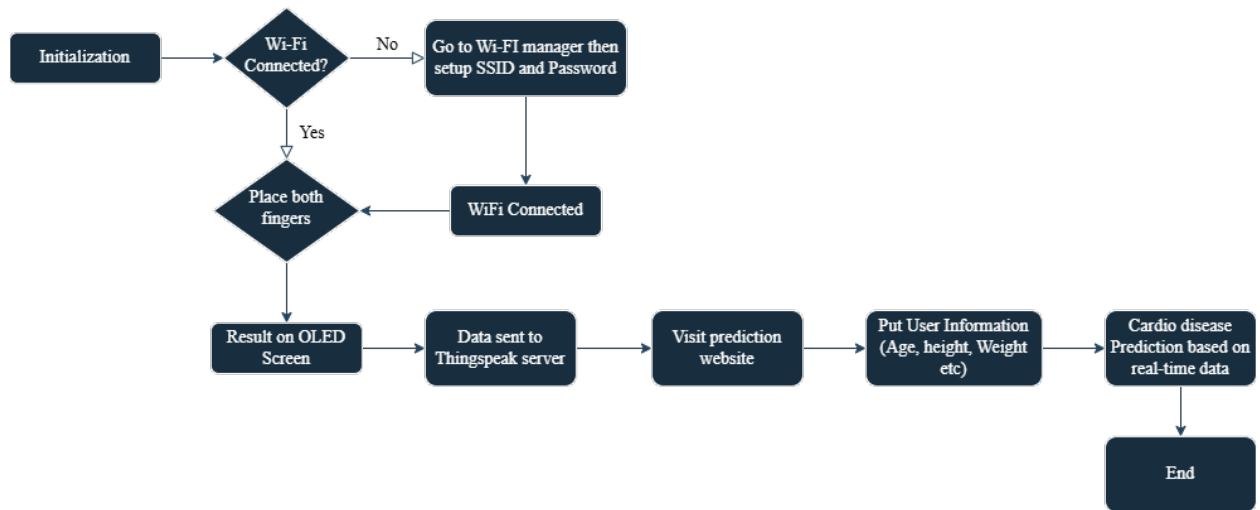


Fig 5: Flowchart of the project from user end

Results:



Fig 6: Measured data on OLED screen

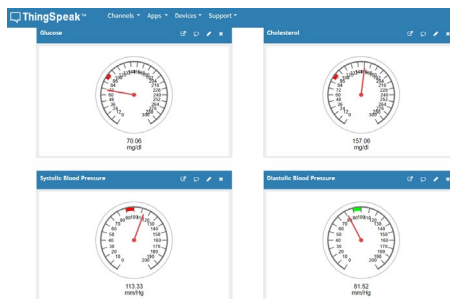


Fig 7: Data sent on ThingSpeak Server



Fig 8: Measured SBP and DBP on ARM Blood pressure device

The form is titled "Heart Disease Prediction System" and contains the following fields and values:

- Age (years): 24
- Gender: Male
- Height (inches): 66
- Weight (kg): 70
- Smoking: No
- Alcohol intake: No
- Physical activity: No
- Glucose (mg/dL): 70.06
- Cholesterol (mg/dL): 157.06
- Systolic BP (ap_hi): 113.33
- Diastolic BP (ap_lo): 81.52

A green "Predict" button is located at the bottom of the form.

Fig 9: Data fetched into website

The result is titled "Heart Disease Prediction Result" and states: "There is a 17.00% chance of having cardiovascular disease."

Sensor Data Used:

- Cholesterol: 157.06 mg/dL
- Glucose: 70.06 mg/dL
- Systolic Blood Pressure (SBP): 113.33 mmHg
- Diastolic Blood Pressure (DBP): 81.52 mmHg

A green "Make another prediction" button is located at the bottom of the result box.

Fig 10: Predicted result based on realtime data

Data Analysis:

Table 3: Average Health Metrics for Different Age Groups and Genders, Based on Three Measurements per Individual

Age	Gender	AVG Glucos	AVG Cho	AVG SBP	AVG DB	G OFFset	C Offset	SBP Offset	DBP offset
24	male	74.61	160.71	96.50	77.22	8.02	11.78	7.31	10.57
38	male	97.75	141.46	103.36	77.27	17.58	48.90	48.78	12.01
20	female	81.67	129.67	99.67	67.33	37.00	12.00	16.00	0.00
24	female	81.92	161.15	109.07	74.64	10.32	24.57	55.65	2.32
42	male	85.61	187.05	86.08	73.16	25.26	36.63	51.95	0.18

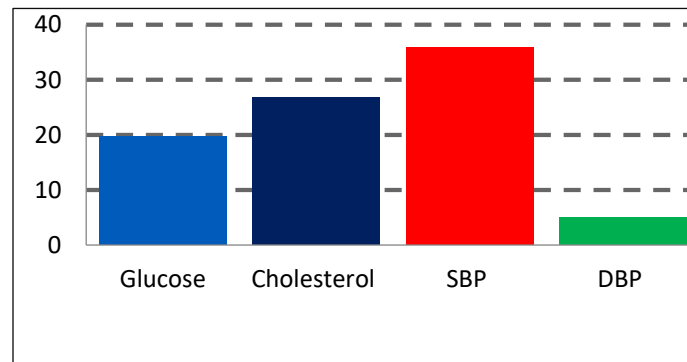


Fig 11: Visualization of average Offset of Health Parameters Across Demographic Groups

Model Analysis:

Age, weight, and blood pressure (systolic and diastolic) are the most significant predictors of cardiovascular disease in this model.

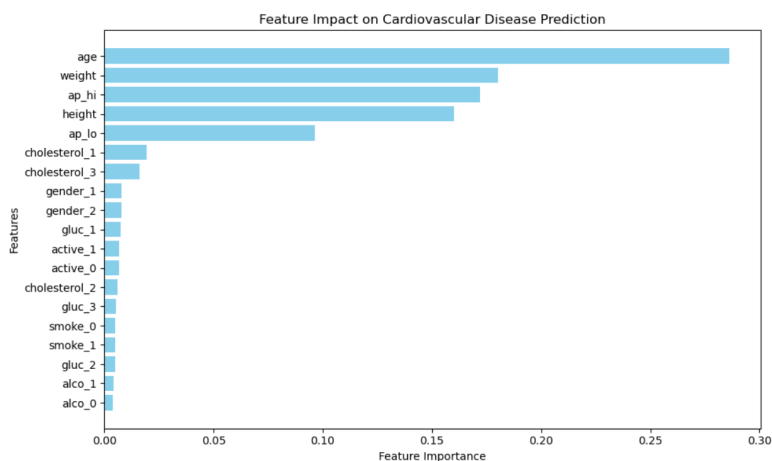


Fig 12: Feature Importance in Cardiovascular Disease Prediction: Age, weight, and blood pressure show the highest impact

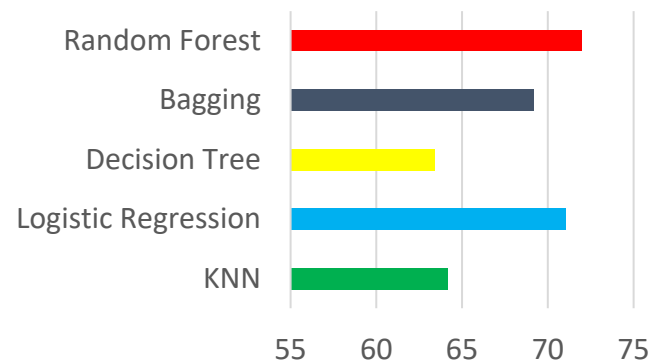


Fig 13: Model Accuracy with various classifiers

Costing:

Table 4: Total cost of the project

Component	Quantity	Price
Sparkfun MAX30101	1	2990/-
SEN SKU0203 pulse sensor	1	320/-
ESP8266 NodeMCU	1	890/-
OLED display	1	280/-
Others	1	200/-
	Total	4680/-

Limitations:

1. Differences in finger skin depth and bone density can lead to measurement inconsistencies.
2. The machine learning model is trained on publicly available data, which may not fully represent the Bangladeshi population.
3. The sensors used are not certified medical devices, potentially limiting accuracy.
4. Health metric calculations are based on external research, not from data obtained specifically for this project.

Future Works:

1. Formulate empirical equations tailored to the device by collecting real-world data in controlled lab settings and validating against certified medical devices.
2. Validate device through clinical trials.

Conclusions:

The *CardioPredictor* project combines non-invasive sensors with machine learning to monitor important health metrics like *blood glucose, cholesterol, systolic, and diastolic blood pressure*. It provides results within 30 seconds and predicts cardiovascular disease risk with approximately 72% accuracy. This project shows how technology can make health monitoring faster and easier, helping with early health intervention. There is room for improvement, but it is a strong step toward creating accessible and reliable health tools.

Acknowledgements:

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